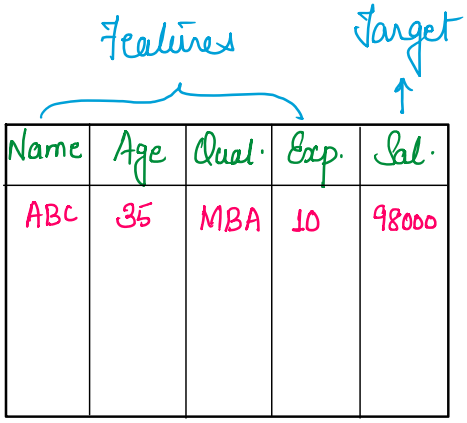
Feature Engineering

Dataset → columns

- i/p (feature)
- o/p (Target)

Ex- Employee Dataset



In ML, we create models (software) that can be used to make predictions.

What to predict (that column) is known as the target.

∴ Target = Mathematical Calculation b/w features

[Target is dependent on features]

[Analyzing the data]

Due to this, we want clean features

Feature Engineering

Transforming or making changes to a feature in order to make it appropriate for analysis or prediction.

* QQ Plot [Quantile-Quantile Plot]:-

- It is a graphical method used to check whether a data follows a particular distribution or not. [most commonly normal distribution]
- This distribution is checked using quantiles/percentiles.

There are two types of quantiles :-

1. Standard Normal Quantile / Theoretical Quantile
2. Sample Quantile

Theoretical Quantile :-

For standard normal distribution, $\mu = 0$, $\sigma = 1$

quantiles are calculated

Percentiles	Theoretical Quantiles
10%	−1.28
25%	−0.67
50%	0
75%	+0.67

Sample Quantile :- Quantiles of our data

Ex- 10, 15, 20, 25, 30

25th percentile → 15
50th percentile → 20
75th percentile → 25

} Sample Quantiles

A graph b/w sample quantiles & theoretical quantiles is drawn

X-axis → Theoretical Quantiles
Y-axis → Sample Quantiles

From the above data → (−0.67, 15)
(0, 20) ----- many more

Using these coordinates, we create a scatterplot

Using these coordinates, we create a scatterplot

- If the points creates a straight line (approx) ⇝ data points are normal (follows a specific dist.)
- If the datapoints does not creates a straight line ⇝ data points are not normal (does not follow a specific dist.)

* Data Transformation :-

Process of converting data from one format to another format.

↳ Why ? → For better analysis, easy interpretation

There are two types of data transformation methods :-

1. Log Transformation
2. Box-Cox Transformation

I. Log Transformation :-

It is a technique where we apply log to our data.

formula →

$$X_{new} = \log(X)$$

Ex- 10, 100, 1000 - - -

applying log → 1, 2, 3, - - - -

appropriate for easy analysis

II. Box-Cox Transformation :-

- Family of transformations
- Automatically decides the best way to transform the data

Formula :-

$$y = \frac{x^{\lambda} - 1}{\lambda}, \quad \lambda \neq 0$$

where, λ = parameter that controls the type of transformation

Advantages of Box-Cox :-

- Reduces the skewness
- Stabilizes variance
- Automatically chooses best transformation

Limitation :-

Box-Cox cannot be applied to -ve data.

★ Feature Scaling :-

It is a process of bringing different features to a similar scale.

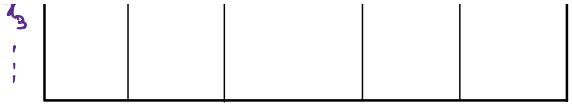
Ex-

	f_1	f_2	f_3	f_4	Target
	Name	Age	Contact	Exp.	Sal.
x_1	ABC	35	9827----	10	98000
x_2	XYZ	29	8265----	6	55000
x_3					
⋮					

for predicting the target

Target = $f_1 + f_2 + f_3 + f_4$

$= 0.75 + 35 + 9827140000 + 10$



Suppose ⇒ ABC = 0.75

= 0.75 + 35 + 9827140000 + 10

= 9827140045.75

↓

Here, contact no. dominates the other columns

In order to avoid such problems, we apply feature scaling

↓

Bring feature values to a similar scale

How to apply feature scaling ?

There are 2 scaling methods :-

1. Min-Max Scaling
2. Standard Scaling